Lecture 3: Stochastic Integration in Hilbert Space

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§3 Stochastic Integration in Hilbert Space

1.3.1 Martingales in Banach space

Let $(E, \|\cdot\|)$ be a (real) Banach space.

Bochner integral (E-valued integration)

Given a finite measure μ on a probability space $(\Omega, \mathcal{A})$, we want to define

$$\int f\, d\mu \;\in\; E.$$

Remark 1. The Bochner integral is the natural generalisation of the Lebesgue integral to functions taking values in a Banach space E. Just as the Lebesgue integral is built by first defining the integral for simple (step) functions and then extending by density, the Bochner integral follows the same strategy but for E-valued functions. The key challenge in infinite dimensions is that one must distinguish between different notions of measurability (strong vs. weak), which coincide in finite-dimensional spaces but can differ in general Banach spaces.

Step functions. For measurable **vector-valued** functions $f\colon \Omega \to E$, given a step function of the form

$$f \;=\; \sum_{k=1}^{n} x_k\, \mathbf{1}_{A_k}, \qquad x_k \in E, \quad A_k \in \mathcal{A}, \quad A_k \cap A_\ell = \emptyset, \tag{$*$}$$

define

$$\int f\, d\mu \;:=\; \sum_{i=1}^{n} \mu(A_k)\, x_k \qquad \text{(linear combination of vectors } x_k\text{)}.$$

Remark 2. The integral of a step function is a finite linear combination of vectors x_k weighted by the measures of the corresponding sets A_k. This is completely analogous to the Lebesgue integral of real-valued simple functions. The requirement that the sets A_k are pairwise disjoint ensures that the definition is unambiguous: each $\omega \in \Omega$ belongs to at most one A_k, so $f(\omega) = x_k$ is uniquely determined.

Bochner inequality. We want to take suitable limits. The **Bochner inequality** gives:

$$\left\| \int f\, d\mu \right\| \;=\; \left\| \sum_{k=1}^{n} \mu(A_k)\, x_k \right\|_E \;\le\; \sum_{k=1}^{n} \|x_k\|_E\, \mu(A_k) \;=\; \int_\Omega \|f\|_E\, d\mu.$$

For disjoint A_k, the norm of f is the real-valued step function:

$$\|f\|(\omega) \;=\; \sum_k \|x_k\| \, \mathbf{1}_{A_k}(\omega),$$

and the right-hand side is just the ordinary **Lebesgue integral** of $\|f\|_E$.

Remark 3. The Bochner inequality is the key estimate that enables extending the integral from step functions to general Bochner-integrable functions. If a sequence of step functions $f_n \to f$ converges in the L^1 norm, then by the Bochner inequality the integrals $\int f_n \, d\mu$ form a Cauchy sequence in E, and since E is complete (being a Banach space), the limit exists and defines $\int f \, d\mu$. This is exactly the same density argument used to extend the Lebesgue integral from simple functions to L^1.

Extension to all integrals. The Bochner inequality allows us to extend $\int f \, d\mu$ to all functions

$$f \colon \Omega \to E \quad \text{satisfying:}$$

- f is **strongly measurable**, i.e.,

  $$f = \lim_{n\to\infty} f_n \qquad \mu\text{-a.e. (in } E),$$

  where f_n are of type $(*)$,

- $\displaystyle\int_\Omega \|f\|_E \, d\mu < \infty.$

Remark 4. In general, it is **not** true that if $f \colon \Omega \to E$ is $\mathcal{A}/\mathcal{B}(E)$-measurable, this implies that f can be approximated by step functions of type $(*)$. Rather, such a function f is only **weakly measurable**, i.e., for any $\ell \in E'$ (= all continuous linear functionals on E), the real-valued map $\omega \mapsto \ell(f(\omega))$ is measurable.

Theorem 1 (Pettis measurability theorem)**.** A necessary and sufficient condition for $f \colon \Omega \to E$ to be **strongly measurable** is that f is **weakly measurable** and **separably valued** (i.e., $f(\Omega) \subseteq E$ is separable).

Remark 5. In particular, if E is separable (e.g. $\mathbb{R}^d$, $L^p(\mathbb{R}^d)$, etc.), then every weakly measurable mapping $f \colon \Omega \to E$ is also **strongly measurable**.

Notation. We write

$$L^p(\Omega, \mathcal{A}, \mu;\, E), \qquad L^p(\Omega, \mathcal{A}, \mu,\, E)$$

for the space of all Bochner-measurable functions $f \colon \Omega \to E$ with $\int_\Omega \|f\|_E^p \, d\mu < \infty$.

Remark 6. The Bochner–Lebesgue spaces $L^p(\Omega, \mathcal{A}, \mu;\, E)$ are the natural framework for random variables and stochastic processes taking values in a Banach space E. For $p = 1$, the space consists of all Bochner-integrable functions. For $p = 2$ and E a Hilbert space, the space $L^2(\Omega, \mathcal{A}, \mu;\, H)$ is again a Hilbert space, which is essential for developing the L^2 theory of stochastic integration.

Properties of Bochner's integral

(i) **Bochner inequality:**

$$\left\| \int f \, d\mu \right\| \;\leq\; \int \|f\| \, d\mu.$$

(ii) **Linearity:** If $L\colon E \to F$ is a **continuous, linear** map (where F is also a Banach space), then

$$L\left(\int f \, d\mu \right) \;=\; \int L(f) \, d\mu.$$

(iii) If $f \in C\big([a,b];\, E\big)$, then

$$f(t) - f(s) \;=\; \int_s^t f'(r) \, dr,$$

where the right-hand side is a **Bochner integral**.

Remark 7. Property (iii) is the **fundamental theorem of calculus** for Banach-space-valued functions. It asserts that if $f\colon [a,b] \to E$ is continuously differentiable, then f can be recovered from its derivative by Bochner integration, exactly as in the scalar case. This result is used extensively in the theory of C_0-semigroups and evolution equations, where solutions are often expressed as integrals of operator-valued functions.

Conditional expectation in Banach spaces

Theorem 2 (Theorem 1.16 — Conditional expectation)**.** Let X be a random variable $X\colon \Omega \to E$ with $X \in L^1(\Omega, \mathcal{A}, P;\, E)$, i.e., X is a Bochner-integrable E-valued random variable. Let $\mathcal{A}_0 \subset \mathcal{A}$ be a sub-σ-algebra of $\mathcal{A}$. Then there exists a P-a.s. uniquely determined

$$X_0 \;\in\; L^1(\Omega, \mathcal{A}, P;\, E)$$

such that

$$\int_A X_0 \, dP \;=\; \int_A X \, dP \qquad \forall\, A \in \mathcal{A}_0.$$

X_0 is called the **conditional expectation** of X given $\mathcal{A}_0$, and we write $X_0 \equiv E(X \mid \mathcal{A}_0)$.

Remark 8. Bochner's inequality implies the following estimate for conditional expectations:

$$\big\| E(X \mid \mathcal{A}_0) \big\|_E \;\leq\; E\big(\|X\|_E \;\big|\; \mathcal{A}_0 \big).$$

In a Hilbert space, it is **not** possible to require monotonicity ($\geq$) for conditional expectations, since there is no natural partial order on H.

Proof (sketch). Start with a step function $X = \sum_{k=1}^n x_k \, \mathbf{1}_{A_k}$ (pairwise disjoint A_k). Then

$$E(X \mid \mathcal{A}_0) \;=\; \sum_{k=1}^n x_k \; E\big(\mathbf{1}_{A_k} \mid \mathcal{A}_0 \big).$$

Hence,

$$\big\| E(X \mid \mathcal{A}_0) \big\| \;\leq\; \sum_{k=1}^n \|x_k\| \; E\big(\mathbf{1}_{A_k} \mid \mathcal{A}_0 \big) \;=\; E\big(\|X\| \mid \mathcal{A}_0 \big).$$

Generalize by using converging sequences.

Remark 9. The extension from step functions to general E-valued random variables follows the same approximation scheme as in the scalar case: approximate X by step functions X_n, apply the conditional expectation to each X_n (which is explicit), and pass to the limit. The Bochner inequality ensures that the conditional expectation operator is a **contraction** on $L^1(\Omega, \mathcal{A}, P; E)$: if $X_n \to X$ in L^1, then $E(X_n \mid \mathcal{A}_0) \to E(X \mid \mathcal{A}_0)$ in L^1, guaranteeing existence and uniqueness of the limit.

Martingales in Banach spaces

Definition 1 (E-valued martingale)**.** An E-valued stochastic process $(M_t)_{t\geq 0}$ is called an $(\mathcal{A}_t)$-**martingale** (with respect to the filtration $(\mathcal{A}_t)$) if:

(i) $E\big(\|M_t\|\big) < \infty$ (Bochner integrability),

(ii) M_t is $\mathcal{A}_t$-adapted,

(iii) $E(M_t \mid \mathcal{A}_s) = M_s, \quad 0 \leq s \leq t.$

Remark 10. The definition directly generalises scalar-valued martingales to the Banach space setting. Condition (i) ensures that the Bochner integral (and hence the conditional expectation in (iii)) is well-defined; it replaces the classical requirement $E(|M_t|) < \infty$. Condition (ii) is the usual adaptedness: M_t may only depend on the information available at time t. Condition (iii) is the **martingale identity**, now interpreted via **Bochner conditional expectation** (Theorem 1.16). The key subtlety in infinite dimensions is that the conditional expectation must be constructed using the Bochner integral theory, and one cannot rely on monotonicity arguments as in the real-valued case.

Remark 11. If (M_t) is an E-valued $(\mathcal{A}_t)$-martingale, then for every $\ell \in E'$ (continuous linear functional), $\big(\ell(M_t)\big)_{t\geq 0}$ is a **real-valued** $(\mathcal{A}_t)$-martingale.

Example: Q-Wiener process. Let (W_t) be a Q-Wiener process taking values in U with filtration $(\mathcal{A}_t)$. Then (W_t) is an $(\mathcal{A}_t)$-martingale, since $E\big(\|W_t\|_U^2\big) < \infty$.

Indeed, using the **canonical representation** of the Wiener process,

$$
\begin{aligned}
E\big(\|W_t\|_U^2\big) &= E\left(\left\|\sum_{k=1}^{\infty} \sqrt{\lambda_k}\, \beta_k(t)\, e_k\right\|_U^2\right) \\
&= E\left(\sum_{k=1}^{\infty} \langle W_t,\, e_k\rangle_U^2\right) \qquad \text{(Parseval's identity)} \\
&= E\left(\sum_{k=1}^{\infty} \lambda_k\, \beta_k^2(t)\right) \\
&\overset{\text{Fubini}}{=} \sum_{k=1}^{\infty} \lambda_k\, \underbrace{E\big(\beta_k^2(t)\big)}_{=\,t} \\
&= t \sum_{k=1}^{\infty} \lambda_k \;=\; t\ \mathrm{tr}(Q).
\end{aligned}
$$

Remark 12. Here (e_k) is a CONS of U consisting of eigenvectors of Q with eigenvalues λ_k, and $(\beta_k(t))$ are independent standard Brownian motions. The interchange of summation and expectation is justified by Fubini's theorem (or monotone convergence), since all terms are non-negative. The identity $E(\beta_k^2(t)) = t$ is the variance of a standard Brownian motion.

Verification of the martingale property. For $s \le t$:

$$E(W_t \mid \mathcal{A}_s) \;=\; E\Big(\underbrace{W_t - W_s}_{\text{independent of } \mathcal{A}_s} + W_s \;\Big|\; \mathcal{A}_s\Big) \;=\; \underbrace{E(W_t - W_s)}_{=0} + W_s \;=\; W_s.$$

Hence the Q-Wiener process is a martingale.

Remark 13. The verification uses two key properties: (1) the increment $W_t - W_s$ is **independent** of $\mathcal{A}_s$ (by definition of the Q-Wiener process), and (2) it has **mean zero** since $W_t - W_s \sim \mathcal{N}(0, (t-s)Q)$. Independence allows us to factor the conditional expectation, and the zero-mean property gives the result. This is the same argument as for scalar Brownian motion, but carried out in the Hilbert space U using the Bochner-integral formulation of conditional expectation.

Doob's maximal inequality

Theorem 3 (Doob's maximal inequality)**.** Let (M_t) be a right-continuous E-valued $(\mathcal{A}_t)$-martingale. Then for $p > 1$:

$$\left(E\left(\sup_{0\le t\le T} \|M_t\|^p\right)\right)^{1/p} \;\le\; \frac{p}{p-1}\,\big(E\big(\|M_T\|^p\big)\big)^{1/p} \qquad \forall\, p > 1.$$

Remark 14. Doob's maximal inequality controls the **supremum process** $\sup_{0\le t\le T} \|M_t\|$ in terms of the **terminal value** $\|M_T\|$. The constant $\frac{p}{p-1}$ diverges as $p \to 1^+$, reflecting the fact that the inequality fails for $p = 1$ (one needs the weaker L^1 maximal inequality in that case). For martingales, the key idea is that $\|M_t\|^p$ is a **submartingale** (by Jensen's inequality applied to the convex function $\|\cdot\|^p$), and the inequality then follows from the classical Doob submartingale inequality. This result is fundamental for proving convergence of stochastic integrals and for establishing pathwise regularity of solutions to SPDEs.

Space of L^p-bounded martingales. Let $\mathcal{M}^p := \big\{(M_t)_{t\in[0,T]} : E$-valued continuous $(\mathcal{A}_t)$-martingale, L^p-bounde
with norm

$$\|M\|_{\mathcal{M}^p} \;:=\; \sup_{0\le t\le T} \big(E\big(\|M_t\|_E^p\big)\big)^{1/p} \;<\; \infty.$$

This is a Banach space for $p > 1$ (Hilbert space if $p = 2$ and E is a Hilbert space).

Remark 15. By Doob's maximal inequality, an equivalent norm on $\mathcal{M}^p$ is

$$\left(E\left(\sup_{0\le t\le T} \|M_t\|_E^p\right)\right)^{1/p}.$$

1.3.2 Stochastic integral

Fix a U-valued Q-Wiener process (W_t) with filtration $(\mathcal{A}_t)$. We want to define

$$\int \Phi(s)\, dW_s \;\in\; H.$$

Remark 16. The construction of the stochastic integral $\int_0^T \Phi_s \, dW_s$ follows the same three-step programme as the classical Itô integral in finite dimensions: (1) define the integral for *elementary* (step) processes by a finite sum; (2) establish an *isometry* (the Wiener–Itô isometry) that controls the L^2 norm of the integral in terms of the integrand; (3) *extend* the integral to all square-integrable predictable integrands by a density/completion argument. The main new feature in infinite dimensions is that the integrand Φ_s is an *operator* from U to H (not a scalar or a vector), and the isometry involves the **Hilbert–Schmidt norm** of $\Phi_s \circ \sqrt{Q}$ rather than the squared absolute value $|\phi_s|^2$ familiar from the scalar case.

Step 1: Integration of elementary processes. An **elementary process** has the form:

$$\Phi(t) \;=\; \sum_{m=0}^{k-1} \Phi_m \, \mathbf{1}_{]t_m,\, t_{m+1}]}(t), \qquad 0 = t_0 < t_1 < \cdots < t_k = T, \tag{$\star$}$$

where each Φ_m is $\mathcal{A}_{t_m}$-measurable and $\Phi_m \colon U \to H$ is a bounded linear operator, i.e., $\Phi_m \in L(U, H)$.

Then define

$$\int_0^T \Phi_s \, dW_s \;:=\; \sum_{m=0}^{k-1} \Phi_m\big(W_{t_{m+1}\wedge t} - W_{t_m \wedge t}\big) \;\in\; H.$$

This yields a **linear mapping**:

$$I \colon \mathcal{E} \;\longrightarrow\; \mathcal{M}^2,$$

where $\mathcal{E}$ denotes the space of all elementary processes of type ($\star$).

Remark 17. Each term $\Phi_m(W_{t_{m+1}\wedge t} - W_{t_m\wedge t})$ in the definition applies the **operator** $\Phi_m \colon U \to H$ to the Wiener increment $W_{t_{m+1}\wedge t} - W_{t_m \wedge t} \in U$, producing an element of H. The adaptedness requirement (Φ_m is $\mathcal{A}_{t_m}$-measurable) ensures that the integrand is determined by information available *before* the increment occurs — this is the **non-anticipativity** condition that distinguishes Itô integration from other stochastic calculi (e.g. Stratonovich).

Step 2: Wiener–Itô isometry.

Remark 18. If Φ takes values in a general Banach space, there is **no** Wiener–Itô isometry. The isometry requires Q to be trace class and $\sqrt{Q}$ to be Hilbert–Schmidt.

Hilbert–Schmidt operators.

Definition 2 (Hilbert–Schmidt operator)**.** An operator $L \in L(U, H)$ is called **Hilbert–Schmidt** if

$$\|L\|^2_{L_2(U,H)} \;:=\; \sum_{k=1}^{\infty} \|L\, e_k\|_H^2 \;<\; \infty$$

for all CONS (e_k) of U.

Remark 19. The definition is independent of the choice of CONS (e_k).

Remark 20. Hilbert–Schmidt operators form a two-sided ideal in the algebra of bounded operators: if $A \in L_2(U, H)$ and $B \in L(H, V)$, $C \in L(V', U)$ are bounded, then $B \circ A$ and $A \circ C$ are also Hilbert–Schmidt. The Hilbert–Schmidt norm satisfies $\|L\|_{\mathrm{op}} \le \|L\|_{L_2(U,H)}$, so every Hilbert–Schmidt operator is bounded, but the converse is false in infinite dimensions. In finite dimensions ($U = \mathbb{R}^n$, $H = \mathbb{R}^m$), every linear operator is Hilbert–Schmidt and $\|L\|^2_{L_2} = \sum_{i,j} L_{ij}^2$ equals the sum of squared matrix entries (the **Frobenius norm**).

Let $L_2(U,H)$ be the (linear) space of all Hilbert–Schmidt operators from U to H. Then $L_2(U,H)$ is again a (real) **Hilbert space** with the scalar product

$$\langle A, B\rangle_{L_2(U,H)} \;:=\; \sum_{k=1}^{\infty} \langle A\, e_k,\; B\, e_k\rangle_H.$$

Wiener–Itô isometry. Then the following **Wiener–Itô isometry** holds:

$$E\left(\left\|\int_0^t \Phi_s\, dW_s\right\|_H^2\right) \;=\; E\left(\int_0^t \left\|\Phi_s \circ \sqrt{Q}\,\right\|_{L_2(U,H)}^2 ds\right).$$

In particular,

$$\left\|\int \Phi_s\, dW_s\right\|_{\mathcal{M}_T^2}^2 \;=\; E\left(\int_0^T \left\|\Phi_s \circ \sqrt{Q}\,\right\|_{L_2(U,H)}^2 ds\right),$$

where the left-hand side is the norm in the space of continuous H-valued, $\mathcal{A}_t$-adapted martingales.

Remark 21. The Wiener–Itô isometry is the infinite-dimensional analogue of the classical Itô isometry $E\big[\big(\int_0^T \phi_s\, dB_s\big)^2\big] = E\big[\int_0^T \phi_s^2\, ds\big]$ for scalar Brownian motion. The key difference is that the integrand Φ_s is now an *operator* from U to H, and the squared integrand $|\phi_s|^2$ is replaced by the **Hilbert–Schmidt norm** $\|\Phi_s \circ \sqrt{Q}\|_{L_2(U,H)}^2$. The appearance of $\sqrt{Q}$ reflects the spatial correlation of the driving noise: the composition $\Phi_s \circ \sqrt{Q}$ "pre-whitens" the noise, and the Hilbert–Schmidt norm measures the total energy injected into H per unit time. This isometry is the foundation for extending the stochastic integral from elementary processes to all square-integrable predictable integrands via a density argument, exactly as the scalar Itô integral is extended.

Proof sketch. We compute, writing $\Delta_m := W_{t_{m+1}} - W_{t_m}$:

$$\begin{aligned} E\left(\left\|\int_0^T \Phi_s\, dW_s\right\|_H^2\right) \;&=\; E\left(\left\|\sum_{m=0}^{k-1} \Phi_m\, \Delta_m\right\|_H^2\right) \\ &=\; \sum_{m,n} E\big(\langle \Phi_m\, \Delta_m,\; \Phi_n\, \Delta_n\rangle_H\big). \end{aligned}$$

Remark 22. For $m \neq n$, the cross terms vanish: $E\big(\langle \Phi_m\, \Delta_m,\; \Phi_n\, \Delta_n\rangle_H\big) = 0$. This follows from the independence of the Wiener increments and the adaptedness of Φ_m.

Hence only the diagonal terms $m = n$ survive. Using the **canonical representation** of the Wiener process, $\Delta_m = \sum_k \sqrt{\lambda_k}\, \big(\beta_k(t_{m+1}) - \beta_k(t_m)\big)\, e_k$, we get:

$$\begin{aligned} E\big(\langle \Phi_m\, \Delta_m,\; \Phi_n\, \Delta_n\rangle_H\big) \;=\; \sum_{k,\ell} \sqrt{\lambda_k}\, \sqrt{\lambda_\ell}\, E\Big[\big\langle \Phi_m(e_k),\; \Phi_n(e_\ell)\big\rangle_H \\ \cdot \big(\beta_k(t_{m+1}) - \beta_k(t_m)\big)\big(\beta_\ell(t_{n+1}) - \beta_\ell(t_n)\big)\Big]. \end{aligned}$$

Remark 23. When $m = n$, the expectation of the product of Brownian increments gives $E\big((\beta_k(t_{m+1}) - \beta_k(t_m))(\beta_\ell(t_{m+1}) - \beta_\ell(t_m))\big) = \delta_{k\ell}\,(t_{m+1} - t_m)$, where $\delta_{k\ell}$ is the Kronecker delta. This is because the β_k are independent standard Brownian motions. The sum then reduces to the Hilbert–Schmidt norm $\|\Phi_m \circ \sqrt{Q}\|_{L_2(U,H)}^2$.

Detailed computation (continued). Case $m < n$: Starting from the cross term,

$$
\begin{aligned}
&E\big(\langle \Phi_m \, \Delta_m, \ \Phi_n \, \Delta_n \rangle_H\big) \\
&= \sum_{k,\ell} \sqrt{\lambda_k} \, \sqrt{\lambda_\ell} \, E\Big(\big\langle \Phi_m(e_k), \ \Phi_n(e_\ell) \big\rangle_H \, \big(\beta_k(t_{m+1}) - \beta_k(t_m)\big) \, \big(\beta_\ell(t_{n+1}) - \beta_\ell(t_n)\big)\Big).
\end{aligned}
$$

Now apply the tower property with respect to $\mathcal{A}_{t_n}$:

$$
\begin{aligned}
&E\Big(\big\langle \Phi_m(e_k), \ \Phi_n(e_\ell) \big\rangle_H \, \big(\beta_k(t_{m+1}) - \beta_k(t_m)\big) \, \big(\beta_\ell(t_{n+1}) - \beta_\ell(t_n)\big)\Big) \\
&= E\Big(E\Big[\big\langle \Phi_m(e_k), \ \Phi_n(e_\ell) \big\rangle_H \, \big(\beta_k(t_{m+1}) - \beta_k(t_m)\big) \, \big(\beta_\ell(t_{n+1}) - \beta_\ell(t_n)\big) \,\Big|\, \mathcal{A}_{t_n}\Big]\Big) \\
&= E\Big(\big\langle \Phi_m(e_k), \ \Phi_n(e_\ell) \big\rangle_H \, \big(\beta_k(t_{m+1}) - \beta_k(t_m)\big) \, E\big[\beta_\ell(t_{n+1}) - \beta_\ell(t_n) \,\big|\, \mathcal{A}_{t_n}\big]\Big) \\
&= 0.
\end{aligned}
$$

Here Φ_m and Φ_n are $\mathcal{A}_{t_n}$-measurable (since $m < n$ and the integrand is adapted), and $\beta_k(t_{m+1}) - \beta_k(t_m)$ is also $\mathcal{A}_{t_n}$-measurable because $t_{m+1} \le t_n$. The future increment $\beta_\ell(t_{n+1}) - \beta_\ell(t_n)$ is independent of $\mathcal{A}_{t_n}$ and has mean zero, so

$$
E\big[\beta_\ell(t_{n+1}) - \beta_\ell(t_n) \mid \mathcal{A}_{t_n}\big] = 0.
$$

Thus every summand is zero, and therefore the whole sum is zero. Similarly for $m > n$ by symmetry.

Remark 24. The vanishing of cross terms ($m \neq n$) is the fundamental **orthogonality property** of the stochastic integral and is the reason the Itô integral is a martingale. The argument uses the **tower property** of conditional expectation: for $m < n$, condition on $\mathcal{A}_{t_n}$ to separate the past increment $\beta_k(t_{m+1}) - \beta_k(t_m)$ (which is $\mathcal{A}_{t_n}$-measurable together with Φ_m and Φ_n) from the future increment $\beta_\ell(t_{n+1}) - \beta_\ell(t_n)$ (which is independent of $\mathcal{A}_{t_n}$ with mean zero). Factoring out the $\mathcal{A}_{t_n}$-measurable parts and using $E[\beta_\ell(t_{n+1}) - \beta_\ell(t_n) \mid \mathcal{A}_{t_n}] = 0$ yields zero. This is the same mechanism that makes the classical scalar Itô integral a martingale.

Case $m = n$:

$$
\begin{aligned}
&\sum_{k,\ell} \sqrt{\lambda_k} \, \sqrt{\lambda_\ell} \, E\Big(\big\langle \Phi_m(e_k), \ \Phi_m(e_\ell) \big\rangle_H \cdot \big(\beta_k(t_{m+1}) - \beta_k(t_m)\big) \big(\beta_\ell(t_{m+1}) - \beta_\ell(t_m)\big)\Big) \\
&= \sum_k \lambda_k \, E\big(\|\Phi_m \, e_k\|_H^2\big) \cdot (t_{m+1} - t_m) \\
&= \sum_k E\big(\big\|\Phi_m\big(\sqrt{\lambda_k} \, e_k\big)\big\|_H^2\big) \cdot (t_{m+1} - t_m) \\
&= E\left(\sum_k \big\|\Phi_m \circ \sqrt{Q} \, e_k\big\|_H^2\right) \cdot (t_{m+1} - t_m) \\
&= E\Big(\big\|\Phi_m \circ \sqrt{Q}\,\big\|_{L_2(U,H)}^2\Big) \cdot (t_{m+1} - t_m).
\end{aligned}
$$

Remark 25. The key identity used above is $\sqrt{Q} \, e_k = \sqrt{\lambda_k} \, e_k$, since (e_k) are eigenvectors of Q with eigenvalues λ_k, and hence eigenvectors of $\sqrt{Q}$ with eigenvalues $\sqrt{\lambda_k}$.

Summing over $m = 0, \ldots, k-1$:

$$\begin{aligned}
\sum_{m,n} E\big(\langle \Phi_m \, \Delta_m, \ \Phi_n \, \Delta_n \rangle_H\big) &= \sum_{m=0}^{k-1} E\Big(\big\| \Phi_m \circ \sqrt{Q} \, \big\|^2_{L_2(U,H)}\Big) \cdot (t_{m+1} - t_m) \\
&= E\bigg(\int_0^T \big\| \Phi_s \circ \sqrt{Q} \, \big\|^2_{L_2(U,H)} \, ds\bigg),
\end{aligned}$$

which completes the proof of the Wiener–Itô isometry. □

Remark 26. The Wiener–Itô isometry, once established for elementary processes, extends to all **predictable** integrands Φ satisfying $E\big(\int_0^T \|\Phi_s \circ \sqrt{Q}\|^2_{L_2(U,H)} \, ds\big) < \infty$. The extension proceeds by a standard **completion argument**: approximate Φ by elementary processes $\Phi^{(n)}$ in the $L^2(\Omega \times [0,T])$ norm. By the isometry, $\int \Phi^{(n)} \, dW$ is a Cauchy sequence in $\mathcal{M}^2$, and since $\mathcal{M}^2$ is complete (as a Banach space), the limit exists and defines $\int \Phi \, dW$. This is completely analogous to how one extends the Lebesgue integral from simple functions to L^1, or the scalar Itô integral from step processes to L^2-integrands.

Adjoint operator and trace identity

Let $A \in L_2(U,H)$. The **adjoint operator** $A^* \colon H \to U$ satisfies $A^* \in L_2(H,U)$ (i.e. A^* maps H back to U).

Remark 27. One can prove that the composition $A^*A \colon U \to U$ is a **finite-trace** operator on U.

Indeed, its trace satisfies:

$$\begin{aligned}
\operatorname{tr}(A^*A) &= \sum_k \langle A^*A \, e_k, \ e_k \rangle_U \\
&= \sum_k \langle A \, e_k, \ A \, e_k \rangle_H \\
&= \sum_k \|A \, e_k\|^2_H \\
&= \|A\|^2_{L_2(U,H)} \qquad \text{(HS norm)}.
\end{aligned}$$

Remark 28. This identity shows that the Hilbert–Schmidt norm of A equals the square root of the trace of A^*A:

$$\|A\|_{L_2(U,H)} = \sqrt{\operatorname{tr}(A^*A)}.$$

In particular, $A \in L_2(U,H)$ if and only if A^*A is a trace-class operator on U. The key step uses the definition of the adjoint: $\langle A^*A \, e_k, e_k \rangle_U = \langle A \, e_k, A \, e_k \rangle_H$, which follows from $\langle A^*y, x \rangle_U = \langle y, Ax \rangle_H$ for all $x \in U$, $y \in H$.